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Explainable AI and Augmented Reality in Transesophageal Echocardiography (TEE) Imaging

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Abstract— This study explores how mocked explainable AI (xAI) decision support system and Augmented Reality (AR) impact clinicians' trust and performance in Transesophageal Echocardiography (TEE) imaging. A cohort of 24 Emergency Medicine clinicians assessed three xAI modalities alongside an AR interface. The study found that detailed xAI explanations gained the highest trust and decision accuracy. AR-based decision support was also viewed as innovative, but potentially distracting. Further studies with larger sample sizes in real clinical settings are required to validate these findings.

Keywords— Transesophageal Echocardiography (TEE), Explainable AI (xAI), Augmented Reality (AR), Clinician Trust

I. INTRODUCTION

Medical imaging has long stood as a cornerstone of healthcare innovation, with Computer Vision (CV) techniques significantly contributing to this evolution. Over the years, CV algorithms have shown remarkable capabilities in detecting and diagnosing a wide array of medical conditions, often matching or even exceeding the expertise of human clinicians in specific tasks[1]. However, the increasing complexity of these algorithms has led to a decline in their interpretability, creating a "black box" dilemma that has raised concerns in clinical settings where understanding the rationale behind a diagnosis is imperative[2][3].

To tackle this interpretability challenge, the spotlight has shifted towards explainable Artificial Intelligence (xAI)[4]. xAI is proposed as relatively new approach to make AI models transparent and comprehensible, thereby ensuring that clinicians can trust and act upon AI-generated insights. In the medical imaging field, xAI aims to offer interpretable explanations for AI-driven diagnoses, thereby bridging the trust gap and enhancing the utility of AI in clinical decision-making [5][6]. While existing literature has primarily focused on the classification or segmentation of images, there is a notable absence of studies investigating how these AI models can be presented to clinicians and how clinicians interact with such AI-based decision support systems.

Ultrasound and Transesophageal Echocardiography (TEE) are pivotal in point-of-care diagnostics, offering real-time imaging crucial for immediate decision-making. However, their complexity demands rapid and accurate interpretation, often a challenge even for experienced clinicians[7]. This underscores the role of xAI-based decision

support systems, which can offer not just accurate but also explainable insights[8]. While existing research has largely focused on the technical capabilities of AI in image classification, our study extends this by examining how clinicians interact with and trust these AI systems. We also conducted a qualitative study using semi-structured interviews to explore clinicians' perceptions of an Augmented Reality (AR)-enhanced, AI-based decision support system for cardiac imaging. AR is an emerging technology that overlays digital information, such as images or data, onto the real world. In the medical field, AR has the potential to improve diagnostic and treatment planning by providing clinicians with real-time, interactive data visualization[9], [10]. For this study, we used semi-structured interviews to capture clinicians' perceptions of the AR interface in comparison to traditional diagnostic tools.

In this study, we sought to answer two key research questions: First, we examined the impact of different xAI modalities, created using the Wizard of Oz approach, on clinicians' levels of trust and their interactions with the decision support systems. Second, we explored how clinicians perceive the integration of AR-enhanced xAI decision support systems into their existing workflow, particularly in comparison to traditional diagnostic methods.

II. METHOD

A. Participants

We recruited a cohort of 24 Emergency Medicine (EM) clinicians. Participants were randomly assigned to one of three groups of xAI modalities described in the next section (II.B). Demographic data, including age, sex, education level, ethnicity, and type of hospital affiliation (academic vs. community), were collected to provide context to the findings. All participants provided informed consent prior to participating in the study. The research design was reviewed and approved by the Kuwait International Hospital IRB committee.

B. xAI Modalities:

In medical imaging, xAI modalities have evolved to offer various forms of interpretability. For this study, we utilized the Wizard of Oz approach, collaborating with 3 EM clinicians and 2 computer scientists, to develop and refine three distinct xAI modalities tailored specifically for TEE tasks (Fig. 1):

- **Segmented Areas (SA):** This modality highlighted key anatomical structures and potential abnormalities within the TEE images, aiding clinicians in rapid identification and diagnosis.
- **Segmented Areas with Annotations (SAA):** Building upon the segmented areas, this modality included textual annotations that provided clinically relevant insights directly integrated into the TEE images. Annotations were designed to emphasize key structures like the Left Ventricle, Right Atrium, and other crucial cardiac components.
- **Segmented Areas with Detailed Descriptions (SAD):** This modality expanded upon the annotations, offering clinicians a more comprehensive understanding of the segmented areas. The detailed descriptions were designed to provide an in-depth context. These descriptions elaborated on the clinical significance of the annotated structures, such as the implications of Left Ventricular Ejection Fraction or the presence of Pericardial Effusion.

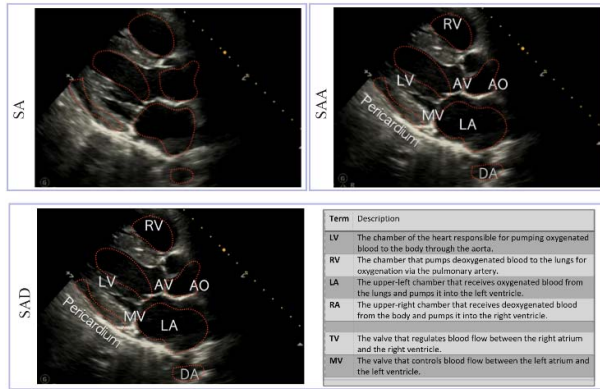


Fig 1. xAI Modalities: Segmented Areas (SA), Segmented Areas with Annotations (SAA), and Segmented Areas with Detailed Descriptions (SAD)

C. AR Implementations

For the AR component of our study, we developed a proof-of-concept application that projected xAI-augmented TEE images onto Microsoft's HoloLens 2. Utilizing Microsoft's built-in libraries, the application was designed to overlay these augmented images within the field of view of the participants. This enabled clinicians to interact with and assess the xAI modalities in a more immersive and integrated manner, simulating a real-world clinical setting.

D. Procedures

Upon completion of a structured training session to acquaint them with both the xAI and AR systems, participants were sequentially introduced to simulated TEE tasks. Each task was designed to mimic real-world scenarios, requiring participants to answer questions regarding TEE images and the support of their assigned xAI modality. Objective performance metrics were recorded for each task.

Upon completion of these tasks, participants were asked to use an AR-based decision support system overlying xAI augmented TEE images to their field of view. In this phase, they were not required to complete any task, instead they were prompted to answer questions regarding their subjective perceptions of AR experience.

E. Tools:

Demographic questionnaires: A form was created to capture a of demographic data including age, sex, years of experience, and familiarity with xAI and AR. **Trust assessment:** The Jian et al (2000) trust questionnaire was employed to collect participants' trust levels in the xAI system[11]. **Perception of social competence:** The Godspeed questionnaire was used to assess participants' perception of the system's social competence[12]. **Performance Metrics:** Decision accuracy and time for each participant were captured.

F. Data Analysis

Quantitative data were first screened for normality using the Shapiro-Wilk test. Based on the distribution, Kruskal-Wallis was employed. Qualitative feedback was coded and thematically analyzed to provide deeper insights into participants' experiences.

III. RESULTS

The study cohort consisted of 24 EM clinicians with an average age of 36 years and a standard deviation of 4.0 years. The gender distribution was 15 males and 9 females. On average, participants had 10 years of experience in EM, with a standard deviation of 2.3 years. The majority were affiliated with academic hospitals (63%). Only 3 participants had prior experience with AR, and the rest had no prior experience with AI, xAI, or AR.

A. Trust Assessment

The Kruskal-Wallis test for trust across the three xAI modalities yielded an H-statistic of 19.31 and a p-value of $< .001$, indicating significant differences among the groups. The mean trust scores were as follows: SA had a mean of 6.84 (SD=0.25), SAA had a mean of 7.4 (SD=0.26), and SAD had a mean of 8.23 (SD=0.17). Pairwise comparisons revealed significant differences between all groups ($< .001$).

B. Godspeed Scores

An analysis of Godspeed scores using the Kruskal-Wallis test yielded significant differences among the three xAI modalities ($H=17.01$, $p<.001$). The SAD modality had the highest mean score of 7.9 (SD=0.27), followed by SAA with a mean of 7.27 (SD=0.32), and SA with a mean of 6.76 (SD=0.31).

C. Task Completion Time

The Kruskal-Wallis test for task completion time indicated an H-statistic of 2.7 and a p-value of .25. The mean task completion times were 54.6 seconds (SD=17.1) for SA, 47.5 seconds (SD=16.8) for SAA, and 68.8 seconds (SD=25.04) for SAD.

D. Decision Accuracy

Significant differences were observed in decision accuracy among the three xAI modalities ($H=6.2$, $p=.04$). The SAD modality led with a mean accuracy of 81.5% (SD=18.9), followed by SAA at 78.2% (SD=17.2), and SA at 67% (SD=14.6). The difference was significant only between SA and SAD, while SA and SAA, as well as SAA and SAD, did not show significant differences in pairwise comparisons.

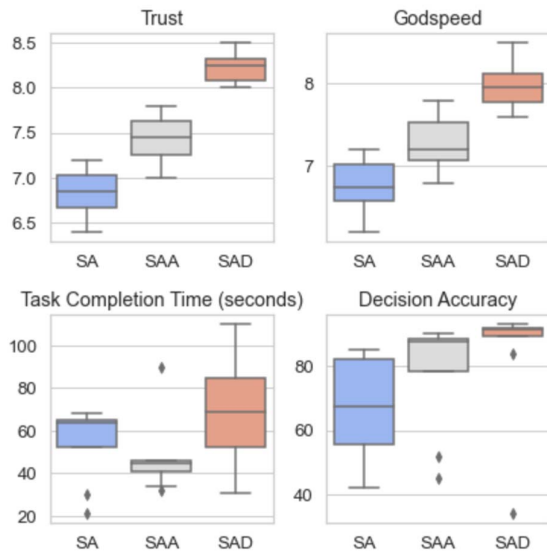


Fig 3. Boxplots comparing key metrics across three xAI modalities (SA, SAA, SAD)

E. Correlation Analysis

A positive correlation was observed between Trust and Godspeed ($r = 0.96$, $p < 0.01$), indicating that higher trust levels were associated with higher Godspeed scores. Trust and Task Completion Time showed a weak positive correlation ($r = 0.12$, $p = .58$), suggesting no significant impact on task completion time. Trust and Decision Accuracy had a moderate positive correlation ($r = 0.36$, $p = .08$), but it was not statistically significant. Similarly, Godspeed and Task Completion Time ($r = 0.08$, $p = .70$) as well as Godspeed and Decision Accuracy ($r = 0.33$, $p = .12$) showed weak to moderate correlations without statistical significance. Lastly, Task Completion Time and Decision Accuracy exhibited a very weak negative correlation ($r = -0.03$, $p = .88$), indicating no significant relationship.

F. Qualitative Analysis

Clinicians, in overall, reported a positive experience with the AR interface. However, some participants noted, "The AR can be distracting in a busy clinical setting, but it's a step in the right direction." Clinicians expressed that the SAD modality was the most helpful. One participant elaborated, "The detailed descriptions really help me understand what I'm looking at, especially when dealing with complex cases." However, some felt that too much information could be overwhelming, suggesting a customizable interface. "Sometimes less is more. I'd like to be able to choose the level of detail that suits me" mentioned by another clinician. Clinicians found that xAI modalities integrated well into their existing workflows. One clinician mentioned, "The AR interface allows for a more dynamic interaction, which is a game-changer. It's like having a second pair of expert eyes". Details of these findings are presented in table 1.

Table 1. Table X: Thematic analysis of clinician perceptions on xAI and AR in TEE imaging

Theme	Description	Example Quotes
Trust in Technology	Clinicians expressed a higher level of trust in xAI modalities that provided more	"I trust it more when I understand it. It's like having a consult on demand." "The more I know, the more I trust it."

	detailed explanations.	
Usability and Workflow	Clinicians found the AR interface to be innovative but expressed concerns about its practicality.	"It's cool but can be distracting. I can see it being more useful as I get used to it." "It's a double-edged sword; innovative but needs getting used to."
Desire for Customization	Clinicians expressed a desire for the ability to customize the level of detail provided.	"I wish I could toggle the details. Sometimes I need more, sometimes I need less." "Customization would be a game-changer."
Transparency	Clinicians valued the transparency offered by detailed xAI modalities.	"Transparency is key in building trust. I want to know why the AI thinks what it thinks." "Transparency is not optional; it's a requirement."

IV. DISCUSSION

The primary objective of this study was to investigate the impact of different mocked xAI modalities on clinicians' trust and their interactions with AI-based decision support systems, in the context of cardiac imaging. We also aimed to explore clinicians' perceptions of an AR-enhanced interface as compared to traditional diagnostic tools.

Our findings show significant differences in trust levels across the three mocked xAI modalities, with the SAD modality engendering the highest trust. This aligns with previous research emphasizing the importance of transparency and detailed explanations in fostering trust in AI systems[13]. Furthermore, our study corroborates the potential of AR for enhancing clinicians' interaction with AI-based decision support systems. This is line with prior studies that have highlighted the utility of AR in medical diagnostics and treatment planning[14].

Our study contributes to the ongoing dialogue about the balance between information utility and potential distraction in AR environments. While AR's innovative approach to data visualization is valued, concerns regarding its distracting nature, as reported by some participants, align with similar findings in earlier research[15]–[17]. This highlights the need for careful design and implementation of AR systems in clinical settings, ensuring that the additional information enhances rather than hinders the decision-making process.

While we observed correlations between trust and other variables like Godspeed scores and decision accuracy, these were not universally significant. This nuanced finding suggests that trust in AI is a multifaceted construct, influenced by various factors including the system's perceived social competence and its actual performance metrics. This complexity has been noted in earlier studies and calls for a more holistic approach to understanding trust in medical AI[18].

One of the primary limitations of this study is the small sample size, which may not be fully representative of the broader population of clinicians. This limitation is compounded by the fact that our xAI modalities were mocked rather than being fully functional systems, which could affect clinicians' responses and the applicability of our findings. Additionally, the study focused solely on cardiac imaging, limiting the generalizability of the findings to other medical specialties. The lack of long-term follow-up also means that

we could not assess the sustainability of trust and usability over time.

Given these limitations, future research should aim to include a more diverse and larger sample of clinicians and extend the study to other medical fields. Longitudinal studies could provide insights into how trust and user experience evolve over time, especially as clinicians become more accustomed to the technology. Further, the development of customizable xAI and AR interfaces could be explored, allowing clinicians to tailor the level of detail and type of information displayed according to their specific needs and expertise.

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